CMR ENGINEERING COLLEGE

UGC AUTONOMOUS

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Kandlakoya (V), Medchal (M), Medchal - Malkajgiri (D)-501401

II-B.TECH I-SEM (R22) II-MID Examinations-November-2025 **Date: 21/11/2025**

Subject: Discrete Mathematics **Time: 10:00 TO 12:00 PM**

Branch: CSE, CSM **Marks: 30 M**

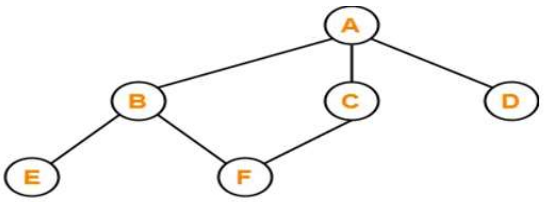
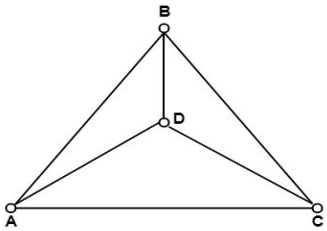
Answer all Questions in Part -A & Answer any FOUR Questions in Part –B

PART-A **5x2=10 M**

1. Define Algebraic System? With Example. [CO3][BTL1]
2. Find the coefficient of $x^3 y^7$ in the expansion of $(x+y)^{10}$. [CO4][BTL3]
3. How many different plates are there that involve 1, 2, or 3 letters followed by 1, 2, 3, or 4 digits? [CO4][BTL3]
4. Define isomorphism of two graphs. [CO5][BTL1]
5. Define graph and planar graph with example? [CO5][BTL2]

PART-B **4x5=20 M**

6. a. Explain semi groups and monoids? [CO3][BTL2]
   b. Draw Hasse diagram representing the positive diviser of 36. [CO3][BTL3]
7. a. Let S be any set and P(S) be its Power set. The partially ordered set (P(S), ⊆) is a lattice.[CO4][BTL4]
   b. Find the middle term(s) of i. $(2x-(1/3x))^{11}$ ii. $((p/x) - (x/p))^9$. [CO4][BTL3]
8. a. Find n if c(n,3): c(n-1,4) = 8: 5. [CO4][BTL3]
   b. A certain computer center employees 100 computer programmers of these 47 can program in Fortan 35 In Pascal 23 can program in both languages. How many can program in neither this 2 languages. [CO4][BTL3]
9. a. Find the number of license plates formed by 3 distinct letters followed by 4 distinct digits with out repetition. [CO4][BTL3]
   b. Explain the Chromatic number of cyclic graph. [CO5][BTL2]
10. Find BFS &DFS of given Tree. [CO5][BTL3]



11. a. Draw the bipartite graphs $K_{2,4}$ and $K_{3,4}$ [CO5][BTL2]
    b. Find the adjacency matrix M_A of graph G shown in Fig [CO5][BTL3]